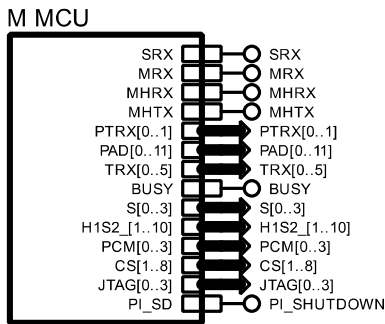
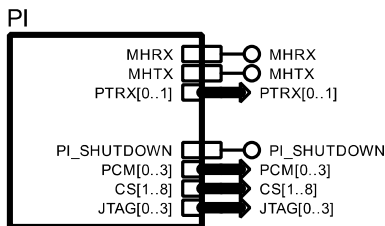
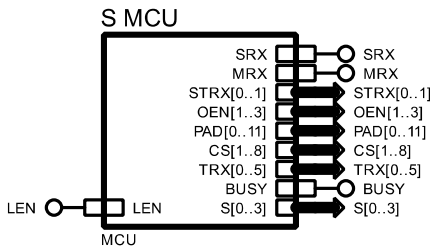




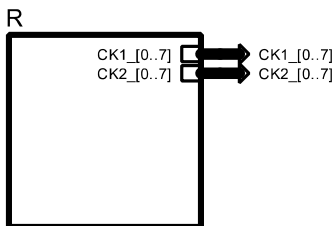
M MCU



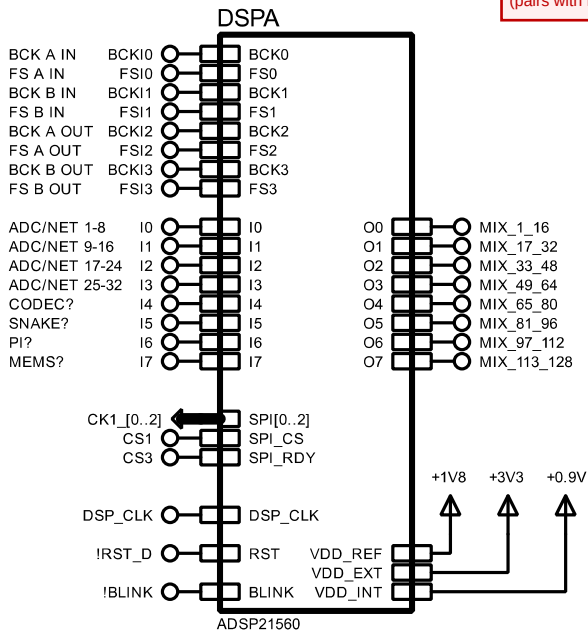
PI



MCU



R

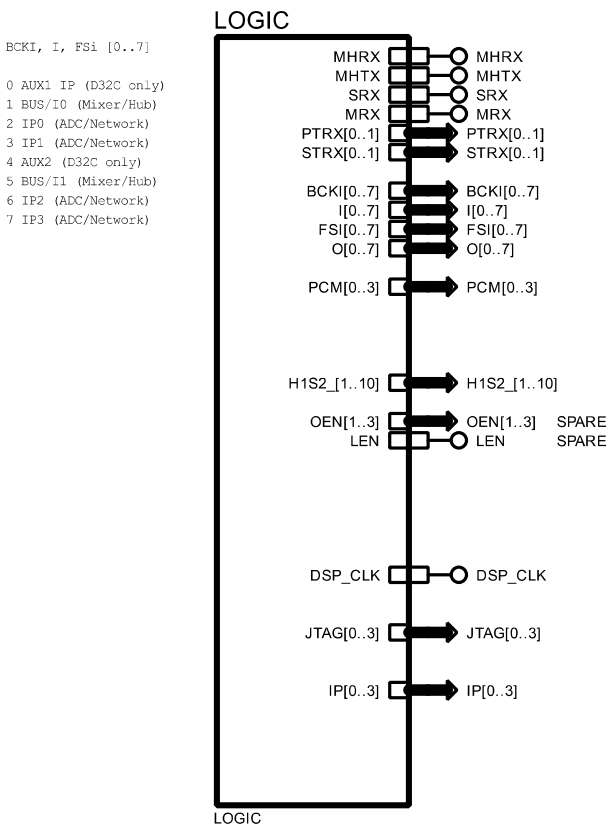


ADSP21560

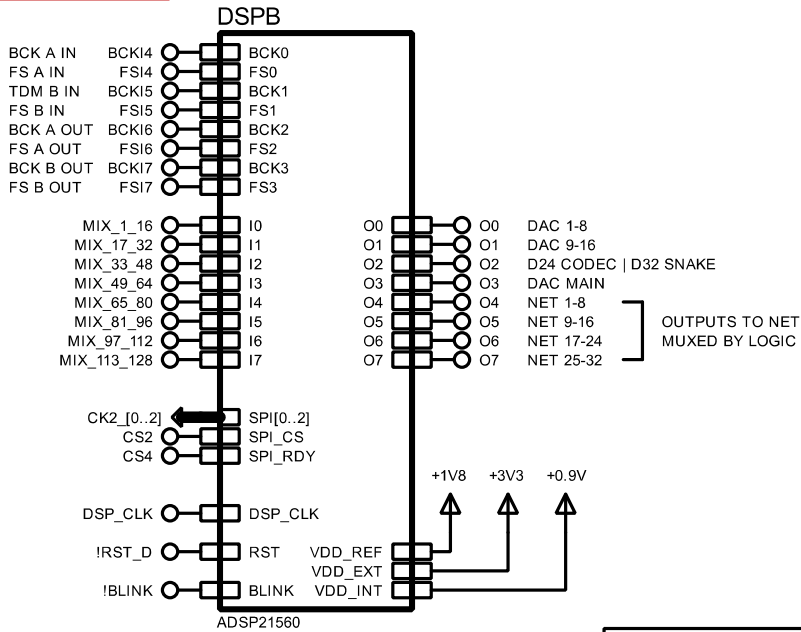
REVIEW: Block labels 'ADSP21560' (below DSPA and DSPB blocks) are stale - detail sheets U5/U6 correctly say ADSP-21564. Update in rev D. Also DSPB left label 'TDM B IN' should read 'BCK B IN' (pairs with FS B IN).

ANALOG RESOLUTION - DSPA (U6) INPUTS (left side of block)
I0/I1/I2 = AD0/1/2: mic/line ch 1-24 from the three ADC8 blocks on D24 Analog rev B (TDM8 via FPC J41/J58, C1/L0 clock pair).
I3 = AD3: NO ADC on D24 Analog - net driven only via D32_COMPAT J33 / LOGIC NET mux ('ADC/NET 25-32' is network-only on D24).
I4 CODEC = AK4916 CODEC4 on ANALOG board (talkback XLR + aux in) -> CDC_O on PLL8_0.
I7 MEMS = surface MEMS mics on switch panels -> LVDS (J12/J13) -> ADAU7302 on Digital bd -> M_BCK/M_FS/M_I2S (PLL7 grp).
I5 snake / I6 PI PCM: digital only.

CROSS-BOARD NOTES (D24 Digital rev C + D24 Analog rev B):
- Converter clocks = C1 (TDM8 BCK 12.288M) + L0 (FS 48k), LOGIC-generated, 33R series on Digital (R111/R112), re-buffered by LVC1G17 (U97/U98) on Analog.
- VERIFY: hardware-map says FPC J41/J42 carry PLL3-6 clock groups; Analog rev B J58/J59 pinouts name only C1/L0 + AD/DA/CDC. Confirm FPC pin alignment (PLL3-6 possibly unused on D24).
- Converter + mic-gain SPI = PI SPI0 branch (R109 on Digital) with CS_C/CS_M via ADC FPC; mic gain via SHIFT daisy-chain on Analog bd.
- hardware-map.md CORRECTION: AK4916 is NOT on the DSP card - it is the AUX_IO 'CODEC4' on the Analog PCBA (PLL8 net naming unchanged).



LOGIC

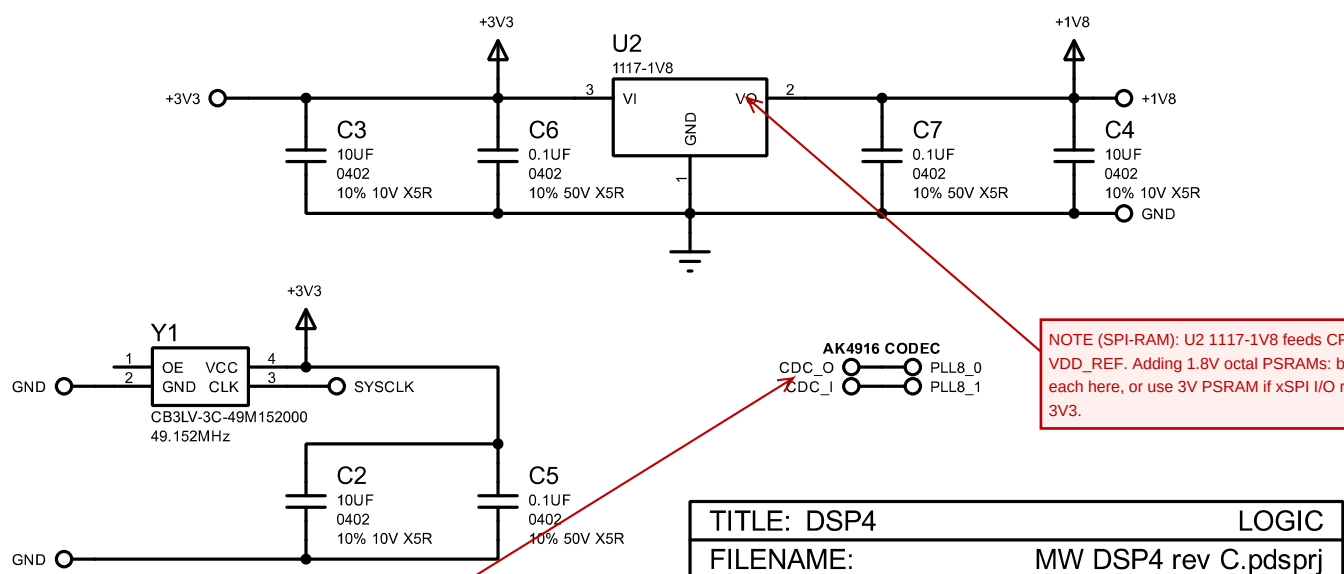
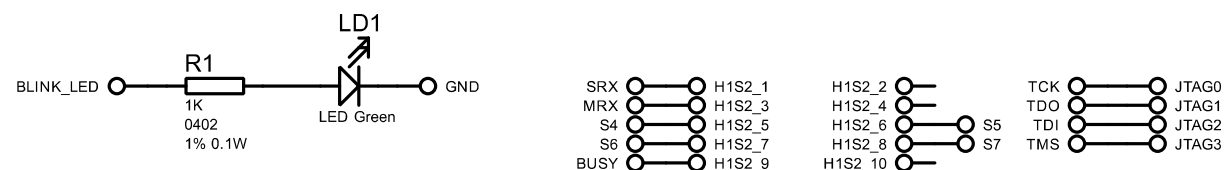
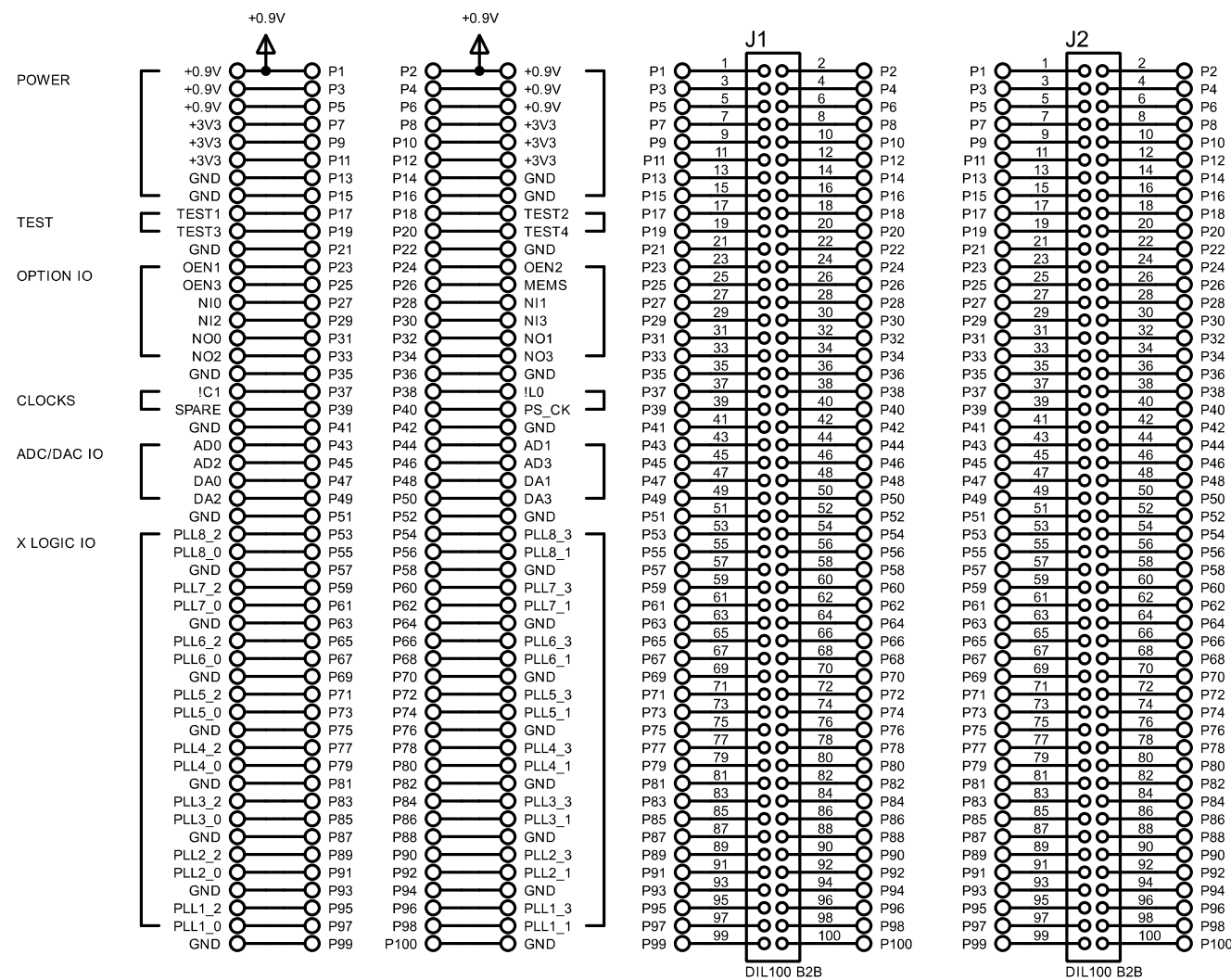
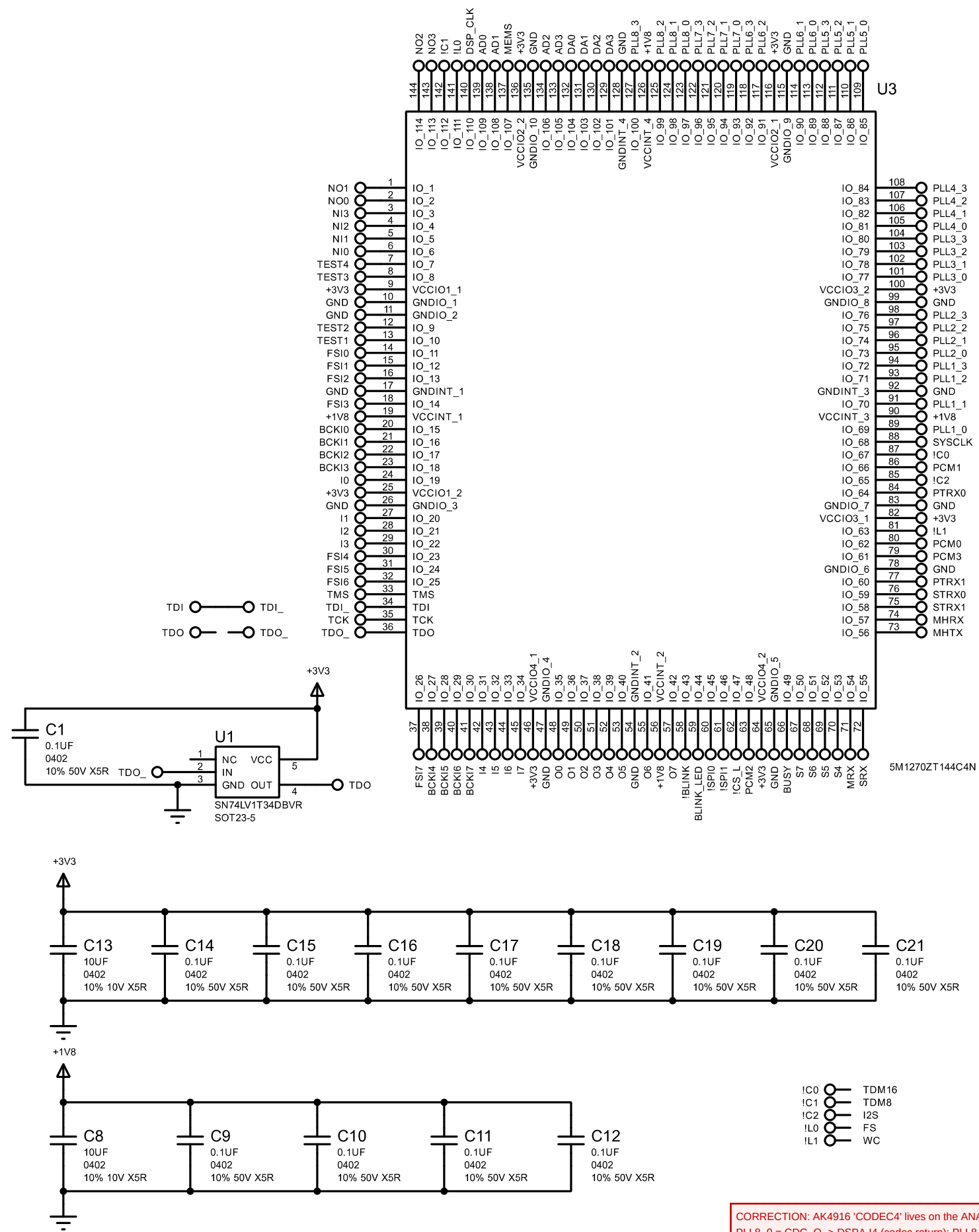


ADSP21560

ANALOG RESOLUTION - DSPB (U5) OUTPUTS (right side of block)
O0 = DA0 -> DAC8 OUT_1-8 (line outs 1-8) via FPC J42/J59.
O1 'DAC 9-16': physical DAC data line is DA3 (OUT_9-16 block input = DA3). DA1 dead-ends at Digital J18 (spare) - LOGIC must route O1 -> DA3.
O2 = CDC_I -> AK4916 codec DAC: talkback speaker feed (SPKR -> TS482 amp on Digital -> panels) + aux out.
O3 'DAC MAIN': NO sink found on D24 Analog rev B (phones amps are fed analog diff pairs) - VERIFY intended target.
O4-O7 = NET to option cards; DA2 -> D32_COMPAT J33 only: digital.

TITLE: DSP4		ROOT	
FILENAME:		MW DSP4 rev C.pdsprj	
AUTHOR:	Peter Watts	PAGE:	1/10
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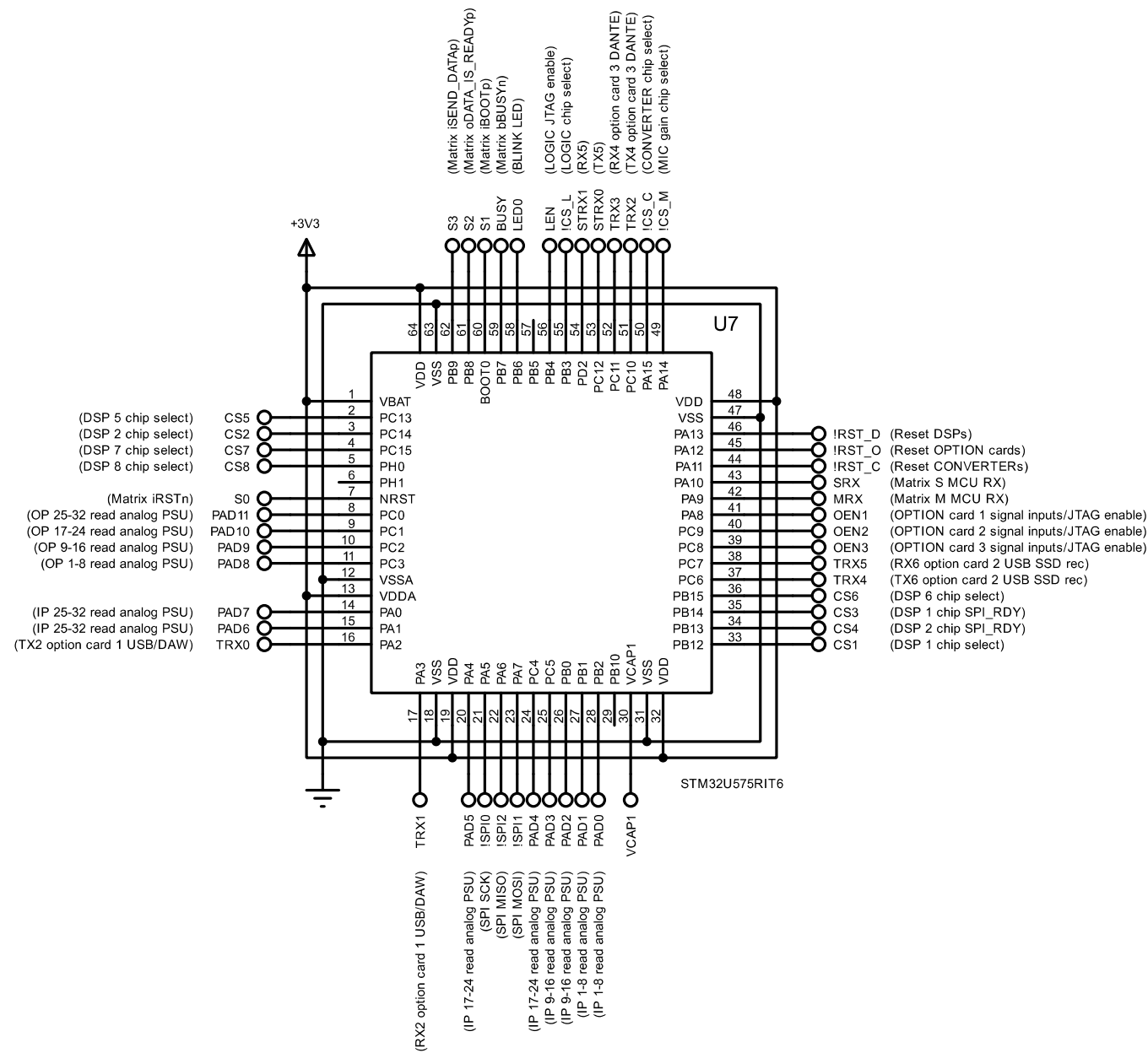
ANALOG BOUNDARY at J1/J2 (X LOGIC IO): AD0-2 = mic ch 1-24 (TDM8 from ADC8s); AD3 = D32-compat/NET only; DA0 = line outs 1-8; DA3 = outs 9-16; DA1 spare (dead-end on Digital); DA2 = D32_COMPAT. C1/L0 = converter BCK/FS pair. All other audio nets here (NI/NO, PCM, MEMS/M_*, PLL1-2) resolve to digital sources only.



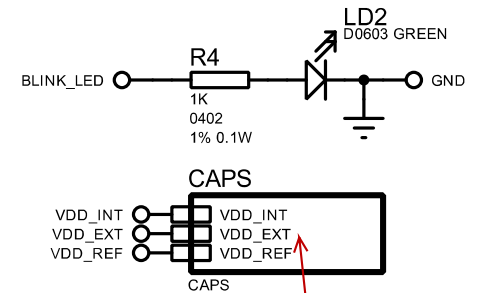
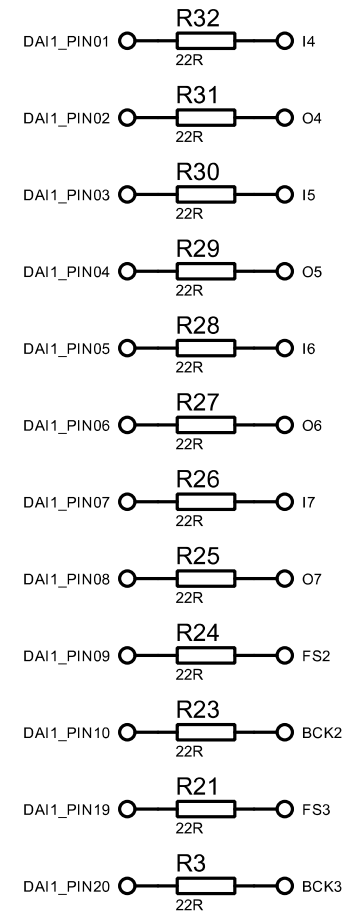
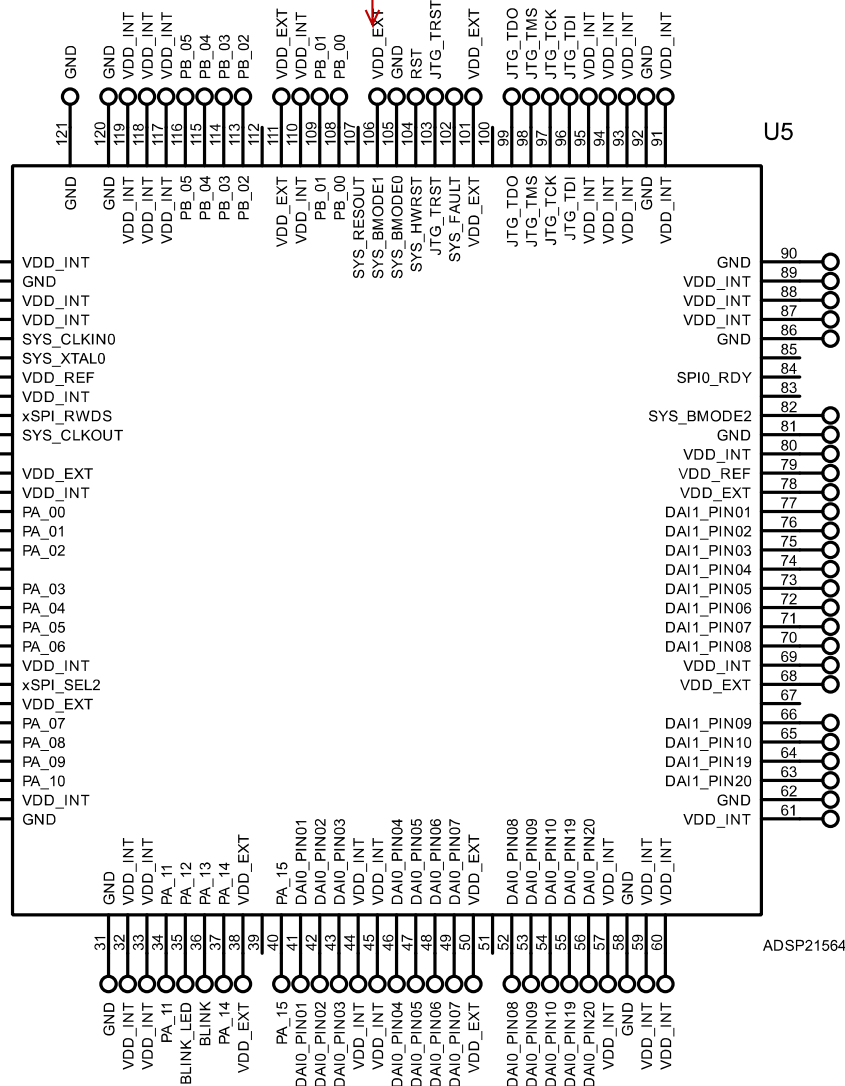
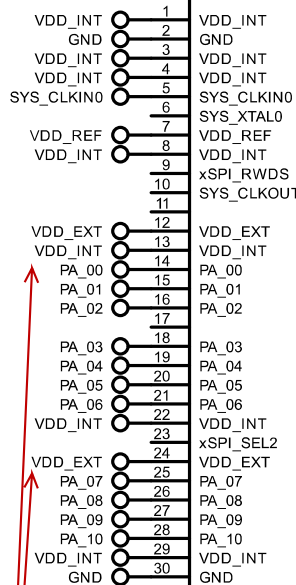
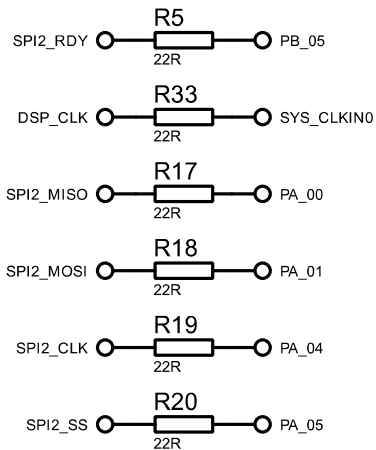
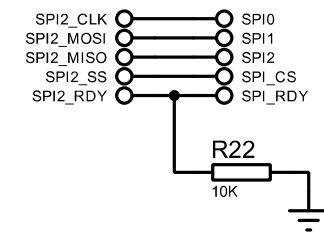
NOTE (SPI-RAM): U2 1117-1V8 feeds CPLD VCCINT + DSP VDD_REF. Adding 1.8V octal PSRAMs: budget ~50-100mA each here, or use 3V PSRAM if xSPI I/O runs on VDD_EXT 3V3.

TITLE: DSP4		LOGIC	
FILENAME:		MW DSP4 rev C.pdsprj	
AUTHOR:	Peter Watts	PAGE:	2/10
DATE:	01/03/2026	REV:	C

CORRECTION: AK4916 'CODEC4' lives on the ANALOG board (AUX_IO block: talkback XLR, aux in, SPKR out) - these stubs are net aliases only.
PLL8_0 = CDC_0 -> DSPA I4 (codec return); PLL8_1 = CDC_1 <- DSPB O2. Update hardware-map.md ('AK4916 codec on DSP card' is wrong).

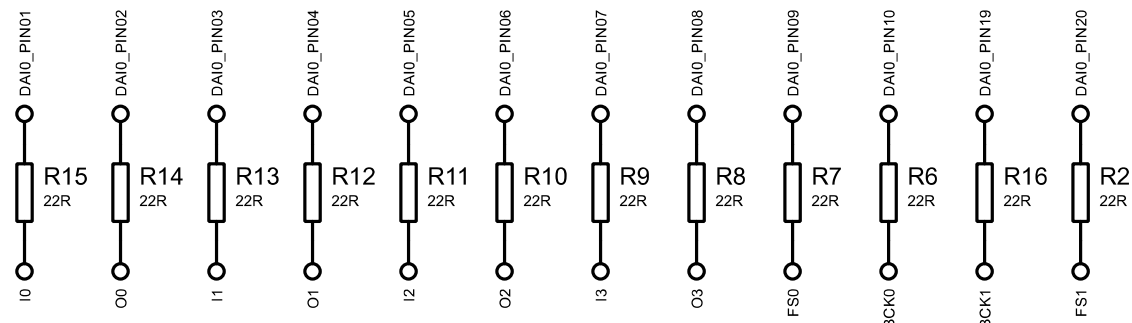


OK: SYS_BMODE[2:0] strapped 0b010 (BMODE2=GND p82, BMODE1=VDD_EXT p106, BMODE0=GND p105) = SPI2 slave boot, matches decision D1 (Pi masters boot). VERIFY: JTG_TRST (p103) appears floating - H1S2 JTAG header carries only TCK/TDO/TDI/TMS. Confirm internal pull, else add 10K to GND.



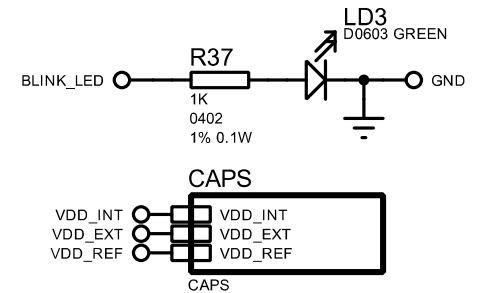
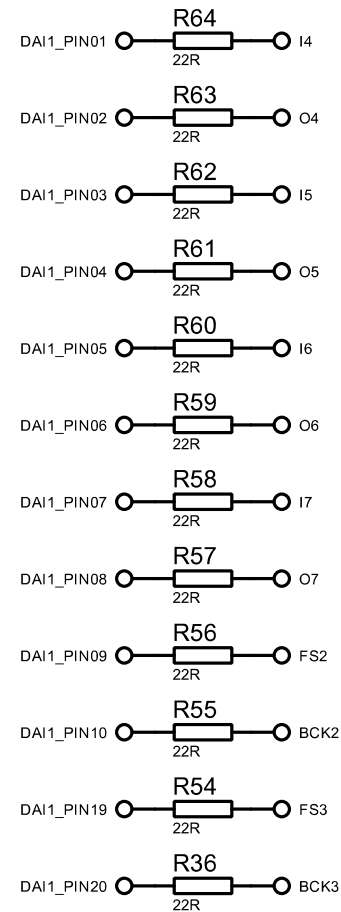
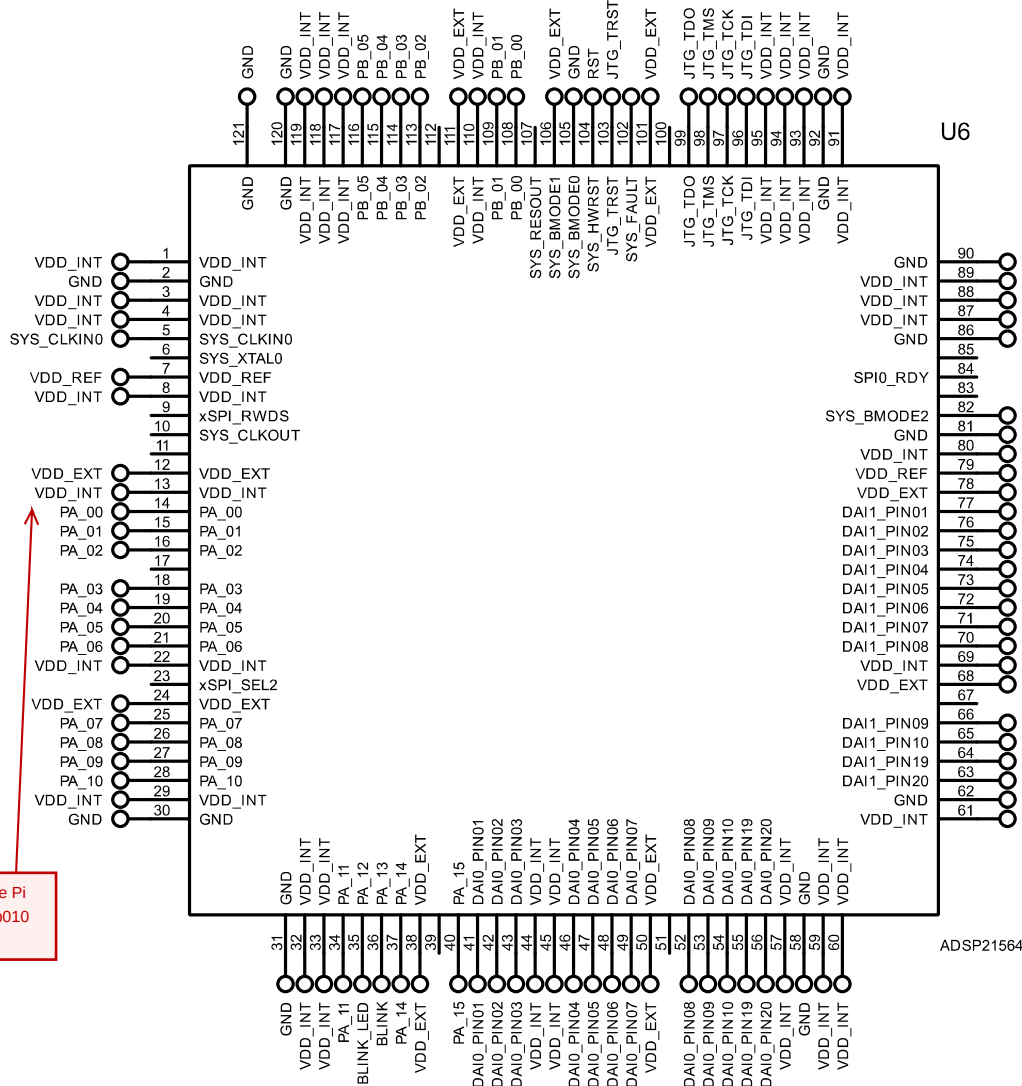
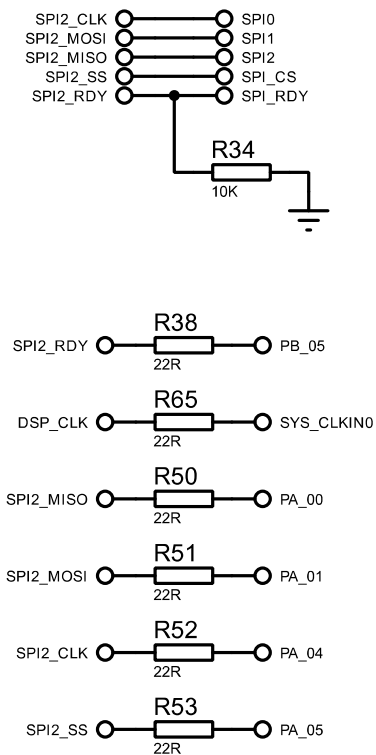
REVIEW / SPI-PRAM option: xSPI_RWDS (p9) and xSPI_SEL2 (p23) are NC and PA_02/03/06/07/08/10 are free, so an octal xSPI PSRAM is pin-feasible on this package. BUT OSPI0 shares the SPI2 / Port-A pin group: the PI link already occupies PA_00/01/04/05 (SPI2 MISO/MOSI/CLK/SEL). Next rev: route PI RUNTIME param link via SPI0/SPI1 and keep SPI2 for slave boot only; verify exact OSPI pin mux + OSPI I/O voltage domain on LQFP (VDD_EXT=3V3 here; 133-200MHz octal PSRAMs are 1.8V - may need 3V-capable PSRAM variant).

REVIEW: 'CAPS' child sheets (pages 9-10) are EMPTY - VDD_INT / VDD_EXT / VDD_REF decoupling for U5/U6 is not captured anywhere. Populate before layout release (cf. LOGIC sheet C8-C21 style arrays).



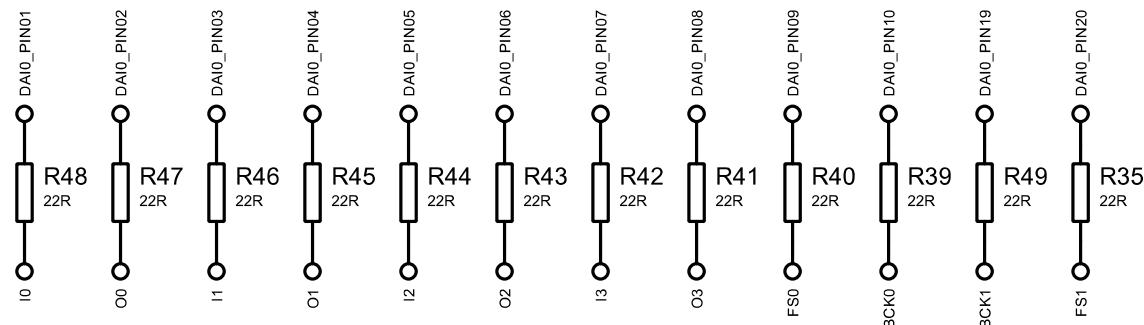
ANALOG RESOLUTION (U5 = DSPB, output DSP): O0 (DAI0 p2) = DA0 -> line outs 1-8; O1 'DAC 9-16' -> physical line is D43 (DA1 is spare); O2 = CDC_I codec (talkback SPKR + aux out); O3 'DAC MAIN' = no sink on D24 Analog rev B (VERIFY); O4-O7 = NET (digital). Inputs I0-I7 = 128 mix buses from DSPA (digital).

TITLE: DSP4		DSPB
FILENAME:		MW DSP4 rev C.pdsprj
AUTHOR:	Peter Watts	PAGE: 4/10
DATE:	01/03/2026	REV: C

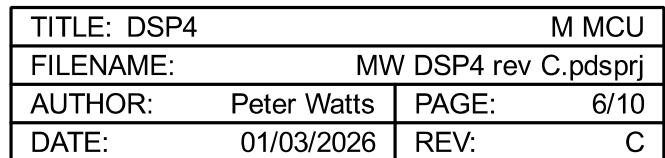
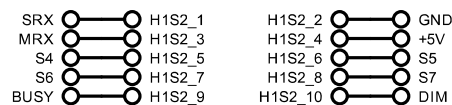


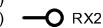
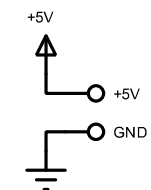
REVIEW: same notes as DSPB sheet (p4/10) apply to U6; OSPI0/xSPI shares Port A with the Pi SPI2 link (PA_00/01/04/05); xSPI_RWDVS/xSPI_SEL2 NC; PA_02/03/06-10 free; BMODE=0b010 SPI2 slave boot OK; CAPS decoupling sheet empty.

ANALOG RESOLUTION (U6 = DSPA, input DSP): I0/I1/I2 (DAIO p1/3/5) = mic ch 1-24 via AD0-2; I3 = no D24 ADC (D32-compat/NET only); I4 (DAI1 p1) = AK4916 talkback/aux ADC via CDC_O; I7 = MEMS surface mics; I5 snake / I6 Pi PCM = digital. Outputs O0-O7 = 128 mix buses to DSPB (digital).



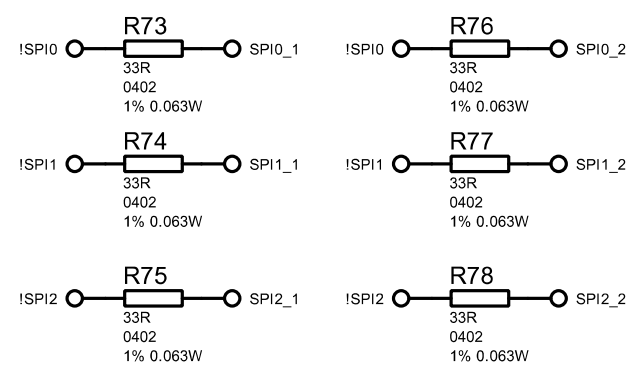
TITLE: DSP4		DSPA	
FILENAME:		MW DSP4 rev C.pdsprj	
AUTHOR:	Peter Watts	PAGE:	5/10
DATE:	01/03/2026	REV:	C



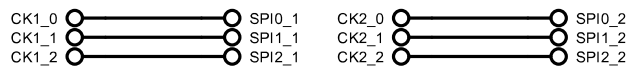


TX2/RX2 only available on CM4.

TITLE: DSP4		PI
FILENAME:		MW DSP4 rev C.pdsprj
AUTHOR:	Peter Watts	PAGE: 7/10
DATE:	01/03/2026	REV: C



REVIEW (SPI-RAM option): both DSPs share ONE Pi SPI bus, split via 33R into CK1/CK2 branches. Because OSPI0 reuses the same DSP Port-A pins, a PSRAM retrofit affects BOTH chips and OSPI clocks (133MHz+ DDR) cannot tolerate these stubs/branches. Next rev: separate runtime param link (SPI0/SPI1) per DSP, keep this bus for boot only.



TITLE: DSP4		R	
FILENAME:		MW DSP4 rev C.pdsprj	
AUTHOR: Peter Watts		PAGE:	8/10
DATE: 01/03/2026		REV:	C

REVIEW: EMPTY SHEET. This is the 'CAPS' child sheet referenced by the CAPS block on the DSPA/DSPB pages - the supply decoupling for the two ADSP-21564s (VDD_INT 0V9, VDD_EXT 3V3, VDD_REF 1V8) is missing from the design. Add bulk + per-pin 100nF arrays before rev D layout/fab.

TITLE: DSP4		CAPS	
FILENAME:		MW DSP4 rev C.pdsprj	
AUTHOR:	Peter Watts	PAGE:	9/10
DATE:	01/03/2026	REV:	C

REVIEW: EMPTY SHEET. This is the 'CAPS' child sheet referenced by the CAPS block on the DSPA/DSPB pages - the supply decoupling for the two ADSP-21564s (VDD_INT 0V9, VDD_EXT 3V3, VDD_REF 1V8) is missing from the design. Add bulk + per-pin 100nF arrays before rev D layout/fab.

TITLE: DSP4		CAPS	
FILENAME:		MW DSP4 rev C.pdsprj	
AUTHOR:	Peter Watts	PAGE:	10/10
DATE:	01/03/2026	REV:	C